SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
ProgramName:B. Tech		**Assignment Type: Lab**	**AcademicYear:**2025-2026
CourseCoordinatorName	Venkataramana Veeramsetty		
Instructor(s)Name	Dr. V. Venkataramana (Co-ordinator) Dr. T. Sampath Kumar Dr. Pramoda Patro Dr. Brij Kishor Tiwari Dr.J.Ravichander Dr. Mohammand Ali Shaik Dr. Anirodh Kumar Mr. S.Naresh Kumar Dr. RAJESH VELPULA Mr. Kundhan Kumar Ms. Ch.Rajitha Mr. M Prakash Mr. B.Raju Intern 1 (Dharma teja) Intern 2 (Sai Prasad) Intern 3 (Sowmya) NS_2 (Mounika)		
CourseCode	24CS002PC215	**CourseTitle**	AI Assisted Coding
Year/Sem	II/I	**Regulation**	R24
Date and Day of Assignment	Week1 - Tuesday	**Time(s)**	
Duration	2 Hours	**Applicableto Batches**	24CSBTB01 To 24CSBTB39
AssignmentNumber:1.2(Present assignment number)/**24**(Total number of assignments)			

Q.No.	Question	*ExpectedTime to complete*
1	Lab 1: Environment Setup – GitHub Copilot and VS Code Integration **Lab Objectives:** ● To install and configure GitHub Copilot in Visual Studio Code. ● To explore AI-assisted code generation using GitHub Copilot.	Week1 - wednesday

- To analyze the accuracy and effectiveness of Copilot's code suggestions.
- To understand prompt-based programming using comments and code context

Lab Outcomes (LOs):
After completing this lab, students will be able to:

- Set up GitHub Copilot in VS Code successfully.
- Use inline comments and context to generate code with Copilot.
- Evaluate AI-generated code for correctness and readability.
- Compare code suggestions based on different prompts and programming styles.

Task Description#1

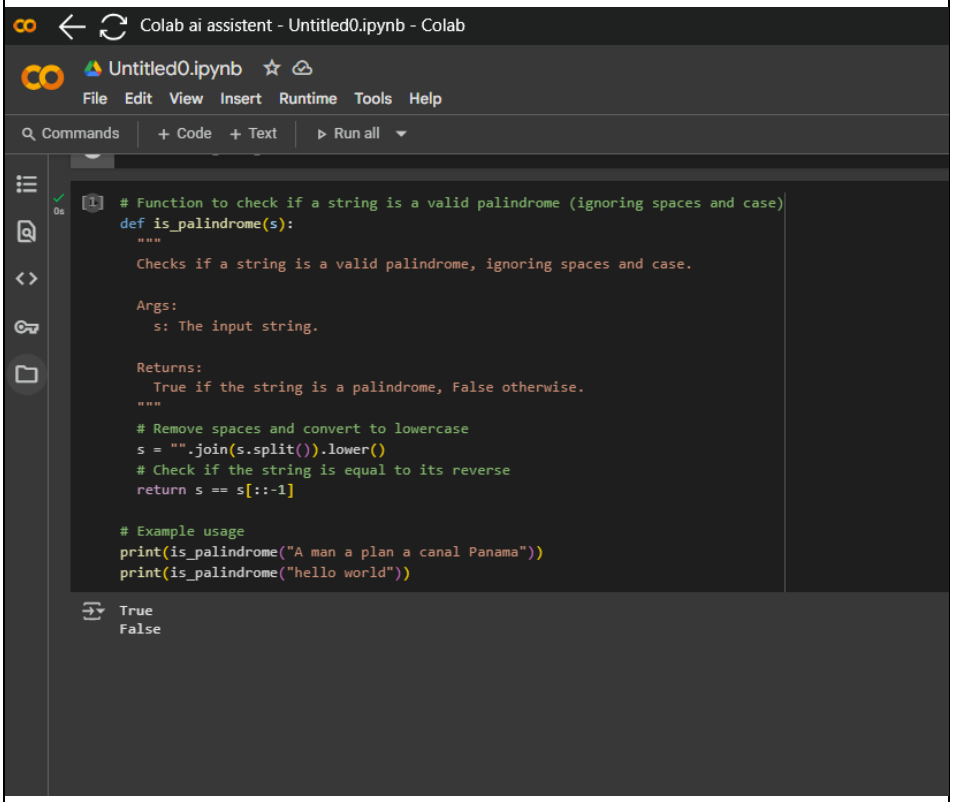
- Write a comment: # Function to check if a string is a valid palindrome (ignoring spaces and case) and allow Copilot to complete it.

Expected Output#1

- A function that correctly returns True for phrases like "A man a plan a canal Panama"

Prompt:

Function to check if a string is a valid palindrome (ignoring spaces and case)



Observation:

the code cell, the is_palindrome function correctly identifies "A man a plan a canal Panama" as a palindrome (True) and "hello world" as not a palindrome (False).

Task Description#2

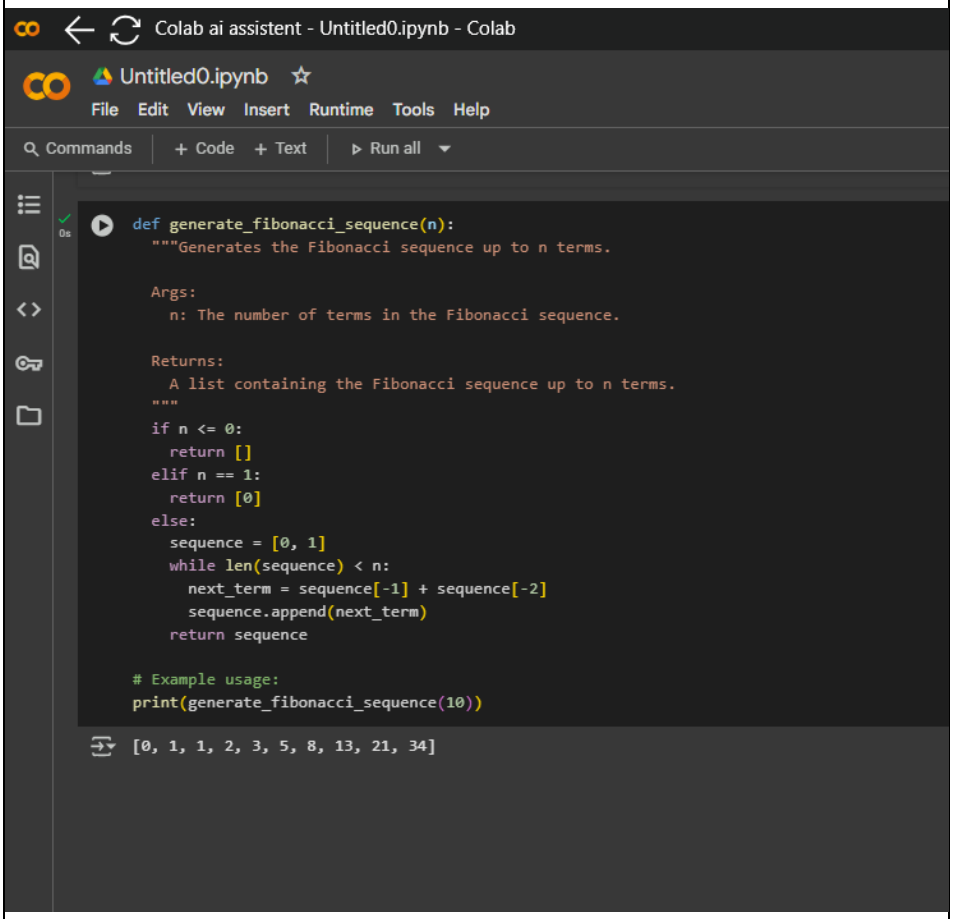
- Generate a Python function that returns the Fibonacci sequence up to n terms. Prompt with only a function header and docstring

Expected Output#2

- AI completes the function logic using loop or recursion with accurate output

Prompt:

function that returns the Fibonacci sequence up to n terms



Obsetvation:

the code cell, the generate_fibonacci_sequence(10) function call successfully generated and printed the first 10 terms of the Fibonacci sequence: [0, 1, 1, 2, 3, 5, 8, 13, 21, 34].

Task Description#3

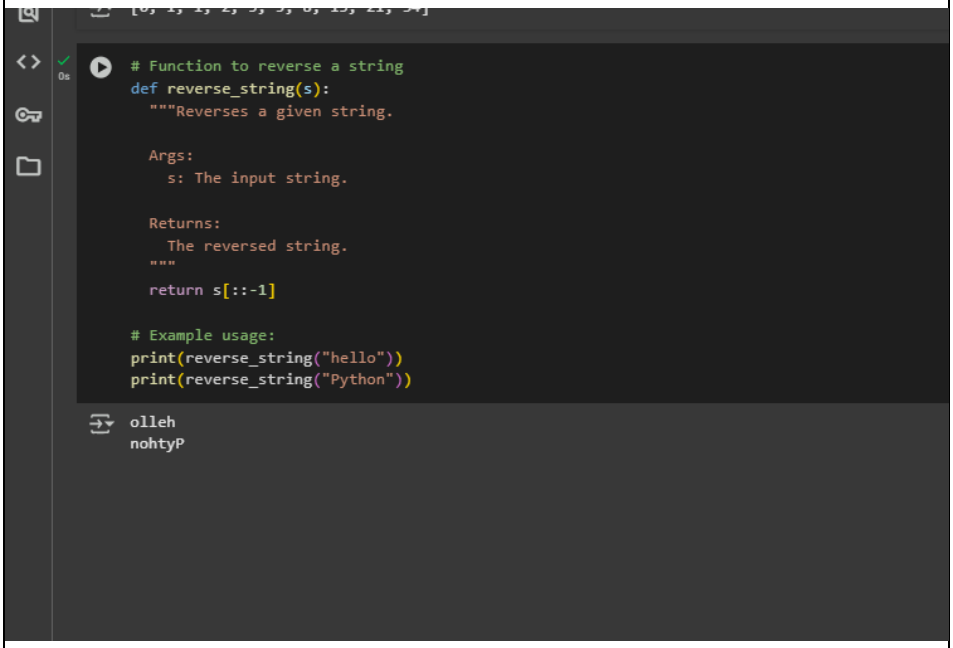
- Write a comment like # Function to reverse a string and use Copilot to generate the function.

Expected Output#3

- Auto-completed reverse function

Prompt:

Function to reverse a string



Observation:

the code cell, the reverse_string function successfully reversed "hello" to "olleh" and "Python" to "nohtyP", as shown in the output.

Task Description#4

- Generate a program that simulates a basic calculator (add, subtract, multiply, divide). Write the comment: # Simple calculator with 4 operations and let AI complete it.

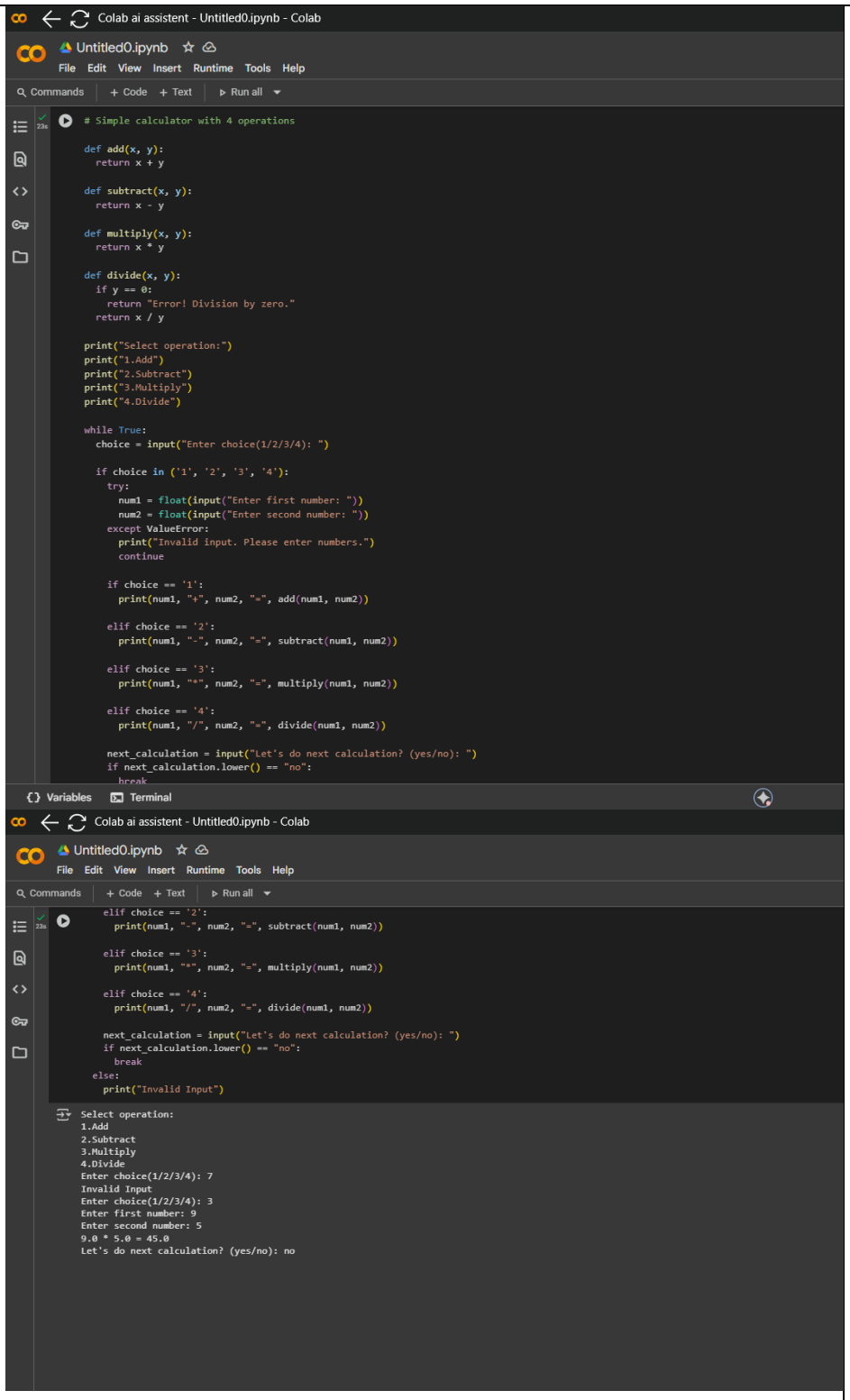
Expected Output#4

- Fully working calculator with input/output and operator selection logic

Prompt:

Simple calculator with 4 operations

a basic calculator (add, subtract, multiply, divide)



Observation:

the code cell, the calculator program successfully took input for an operation (3 for multiply) and two numbers (9 and 5). It then performed the multiplication and printed the result: 9.0 * 5.0 = 45.0. The user then chose to stop the calculation.

Task Description#5

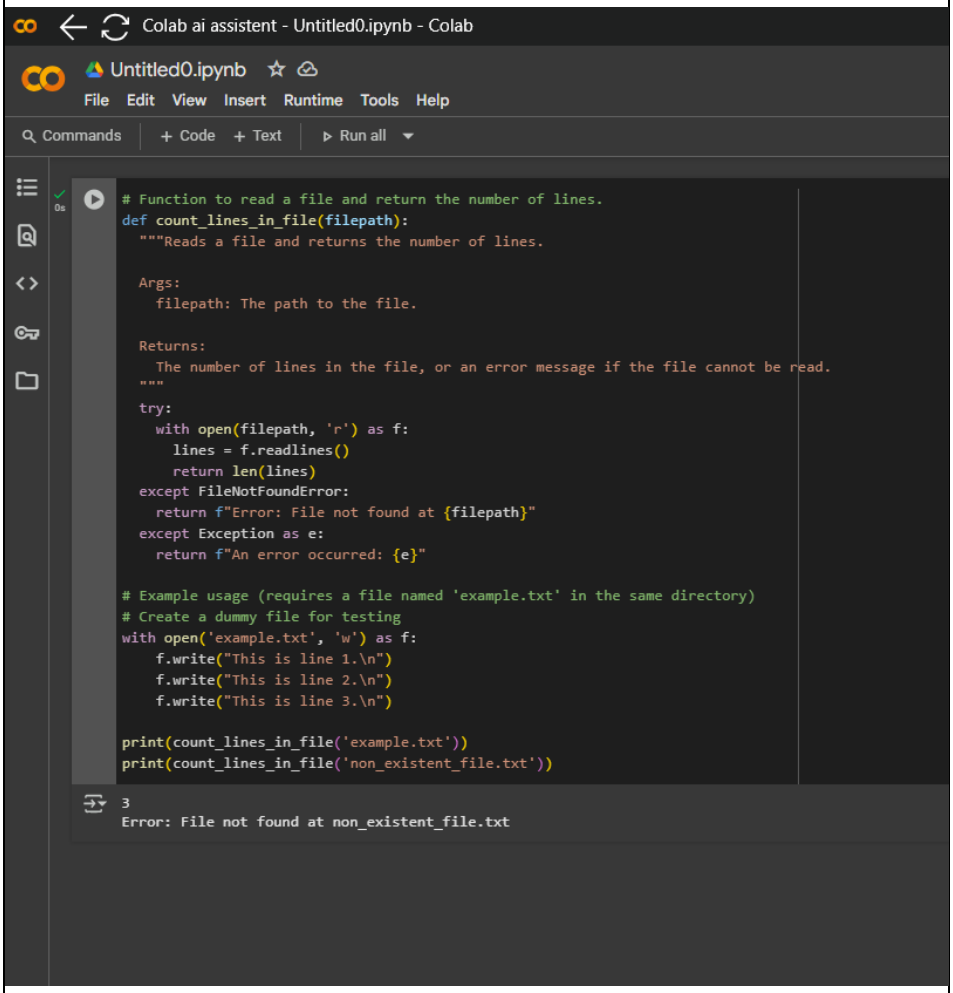
- Use a comment to instruct AI to write a function that reads a file and returns the number of lines..

Expected Output#5

- Functional implementation using open() or with open() and readlines()

Prompt:

write a function that reads a file and returns the number of lines..



Observation:

the code cell, the count_lines_in_file function successfully counted 3 lines in 'example.txt'. When called with 'non_existent_file.txt', it correctly returned the error message "Error: File not found at non_existent_file.txt".

Note: Report should be submitted a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots

Evaluation Criteria:

Criteria	Max Marks
Task #1	0.5
Task #2	0.5
Task #3	0.5
Task #4	0.5
Task #5	0.5
Total	**2.5 Marks**